

## \* Revision:

① Multiple LR: multiple feat.

② overfitting & underfitting:

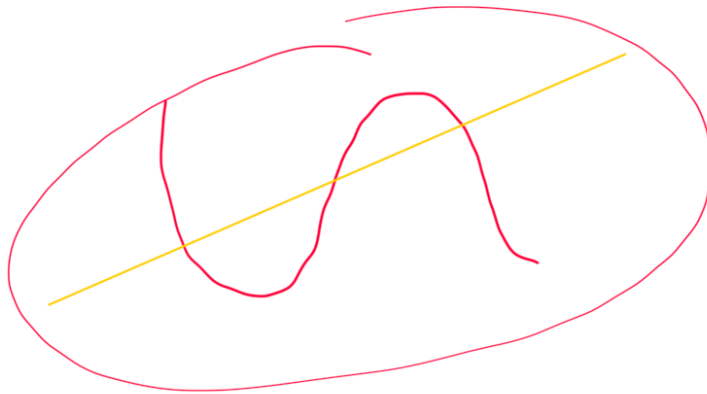
g: if training error  $\downarrow$  & testing  $\uparrow$ : over

if tr error  $\uparrow$  & testing  $\uparrow$ : Under

if tr & te  $\downarrow$ : Perfect

very less data: over

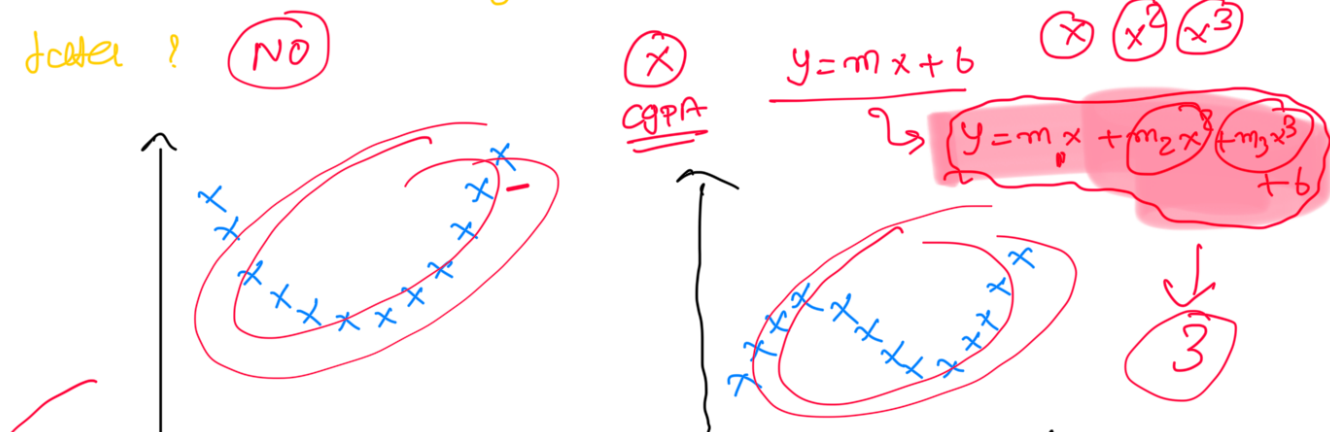
very simple model: underfitting



## \* today's class:

① Polynomial Regression:

Does we have always a linear / linear sort of data? NO



→ To Fit model into this pattern / shape we require to add some non linearity.



⇒ Now, can we convert  $x$  into  $x^1, x^2, x^3 \dots x^n$ ?

$$\underline{y = mx + b} \Rightarrow \underline{y = m_1x + m_2x^2 + m_3x^3 + \dots m_nx^n + b}$$

⇒ Power of  $x$ : (degree) (highest)

degree is hyperparameter

↓  
We have to decide it.

|                     |                                           |
|---------------------|-------------------------------------------|
| $x^2 \rightarrow 2$ | $y = m_1x + m_2x^2 + b$                   |
| $x^3 \rightarrow 3$ | $y = m_1x + m_2x^2 + m_3x^3 + b$          |
| $x^4 \rightarrow 4$ | $y = m_1x + m_2x^2 + m_3x^3 + m_4x^4 + b$ |

⇒ How to decide degree(n):

Go to Visualisation tool. : As degree Increase the curve can bend more & more.

→ Stop before overfitting: (?)

A degree for which we get lowest test Error.  
(Lowest error on unseen data)

low degree : ~~Underfitting~~  
high : Overfitting!  
perfect degree btwn low & high .

⇒ What is poly Reg.?

Polynomial Reg. is a special case of L.R. where features are transformed into polynomial terms to capture non linearity.

$y = mx + b$        $d=3$   
Poly

$y = m_1x + m_2x^2 + m_3x^3 + b$

|       |               |       |
|-------|---------------|-------|
| $x$   | $\rightarrow$ | $x_1$ |
| $x^2$ | $\rightarrow$ | $x_2$ |
| $x^3$ | $\rightarrow$ | $x_3$ |

MLR

$y = m_1x_1 + m_2x_2 + m_3x_3 + b$

Does it look like MLR? YES

So we train it as MLR.

⇒ Go To Kaggle Note book.

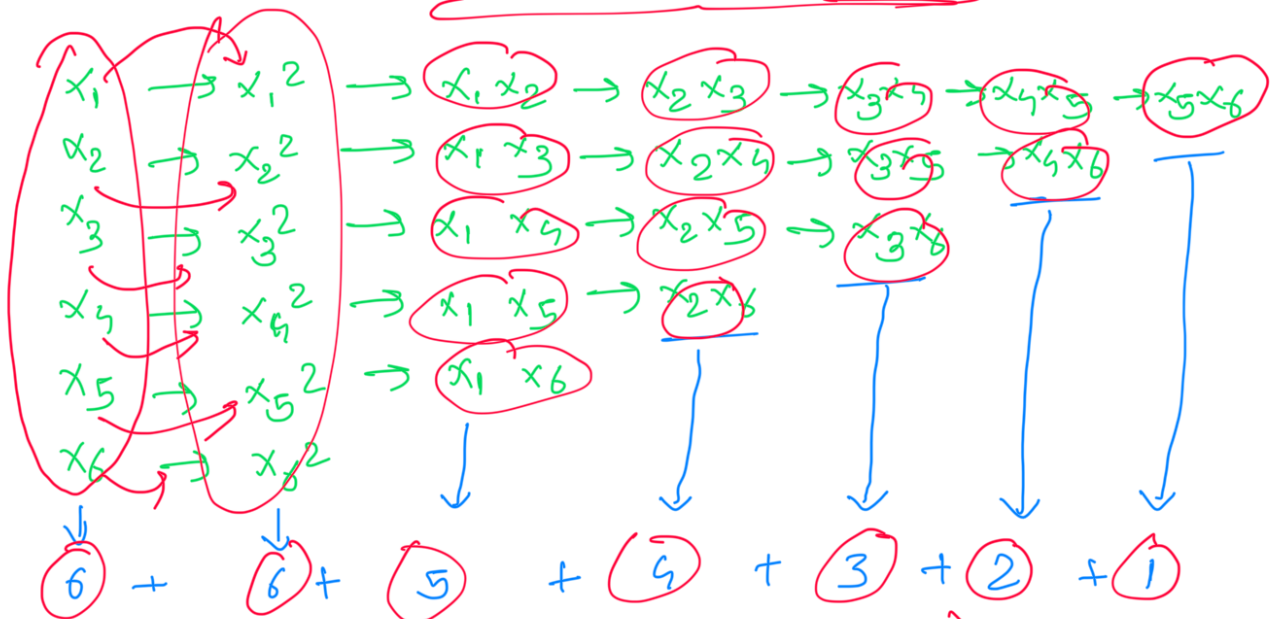
1<sup>st</sup>: training as MLR

2<sup>nd</sup>:  $d=2$  &  $d=3$  Results

3<sup>rd</sup>: transformation of features

degree=2    og features: 6     $\xrightarrow[\text{poly}]{d=2}$     28

$$y = m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4 + m_5 x_5 + m_6 x_6 + b$$
$$\rightarrow y = m_1 x_1 + m_2 x_2 + \dots + m_{27} x_{27} + b$$

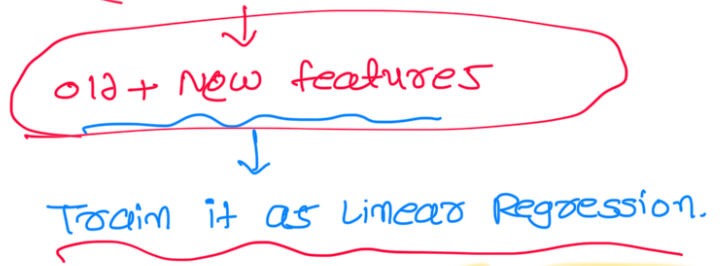


Curse of dim.

$$= 27 + 1 \text{ bias.}$$

- 888...

training of poly: transform features into poly. features.

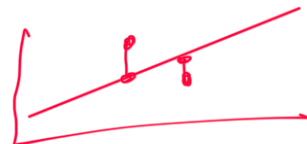


Note: Before training of model using poly features we apply feature scaling.

\* Regression Metrics:  $SE$ ,  $AE$ ,  $\overline{JSE}$

$\underline{\text{MAE:}}$   $\frac{1}{n} \sum (y - \hat{y})$        $\underline{\text{MSE:}}$   $\frac{\sum (y - \hat{y})^2}{n}$        $\underline{\text{RMSE:}}$   $\sqrt{\text{MSE}}$

Why  $V_m$  : (why mean / Avg Error ? why not total Error ?)



Q6: 5 data points  $\xrightarrow[\text{prediction}]{} \text{same} \rightarrow$  10 data points

$$\begin{aligned} b_1 &= 1 & E_6 &= 1 \\ E_2 &= 2 & E_7 &= 1 \\ E_3 &= 1 & E_8 &= 1 \\ E_4 &= 3 & E_9 &= 2 \\ E_5 &= 1 & E_{10} &= 1 \end{aligned}$$

total = 14

$$\text{total} = \frac{15}{10} = 1.5$$

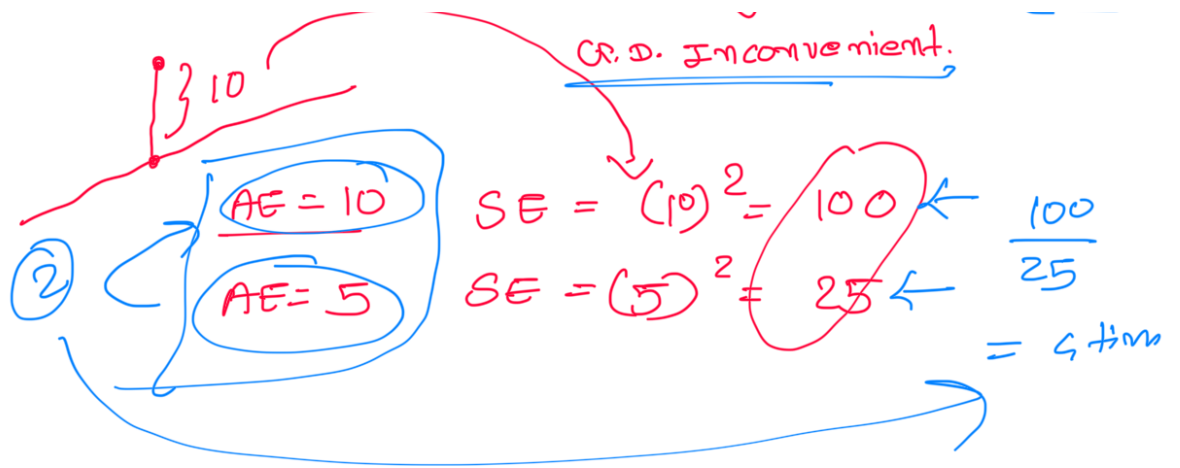
2nd Ex: Training Error (80%) & Testing Error (20%)

1102

Joseph Backus:

Answers:

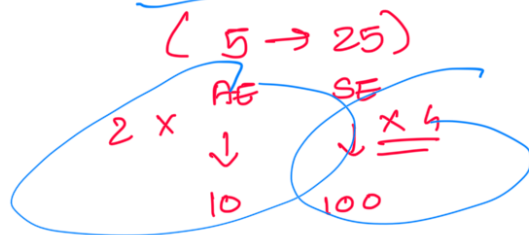
- Not differentiable at zero.  
(bcz of Absolute value function)
- That's why it makes standard



② MSE: CONST  $\frac{(y - \hat{y})^2}{n}$  (IPA  $\rightarrow$  (IPA)<sup>2</sup>)

- We can use it as loss function with G.D.

- Not same unit.
- Not Robust to outliers.



- very sensity to outliers.

③ RMSE:  $\sqrt{MSE}$

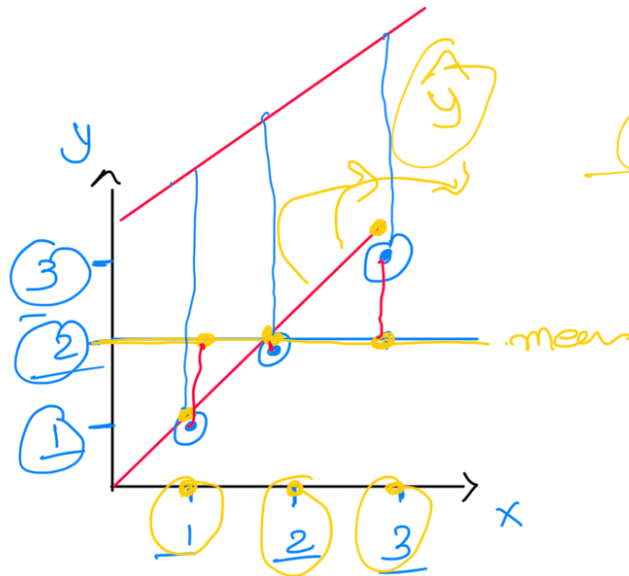
$AE$   
 $IPA$   
Same unit

$SE$   
 $(IPA)^2$

$RSE$   
 $\sqrt{(IPA)^2} = IPA$

Not Robust to outliers.

④  $R^2$  score:



$$\frac{(\sum y)}{n} =$$

$\left\{ \begin{matrix} x, y \\ (1, 1) \\ (2, 2) \end{matrix} \right\}$

$\left\{ \begin{matrix} x, y \\ (3, 3) \end{matrix} \right\}$

Avg of  $y = \frac{1+2+3}{3} = \boxed{2}$

| $x$ | $\hat{y}$ | $\bar{y}$ | $\sum (y - \hat{y})^2$ | $\sum (y - \bar{y})^2$ |
|-----|-----------|-----------|------------------------|------------------------|
| 1   | 1.2       | 2         | 0.2                    | 1                      |
| 2   | 2.2       | 2         | 0.2                    | 0                      |
| 3   | 3.3       | 2         | 0.3                    | 1                      |
|     |           |           | 0.7                    | 2                      |

→  $R^2$  basically compares our model against a mean prediction base-line.

$$R^2 = 1 - \frac{SSR}{SST}$$

$(y - \hat{y})^2 \rightarrow SE$   
 $(y - \bar{y})^2 \rightarrow SST$   
 $SSR \rightarrow$  Error from our model  
 $SST \rightarrow$  Error from mean line

$y = \text{AvgTime}$

SSR = Sum of Squared Error on Regression line

SSM = " " " " " on mean line

$$R^2 = 1 - \frac{SSR}{SSM}$$

As jitno kam Hoga  
utano Hamaara  
 $\frac{SSR}{SSM}$  kam Hoga.

SSM ko ham  
kam nahi kar skate. It will be constant.

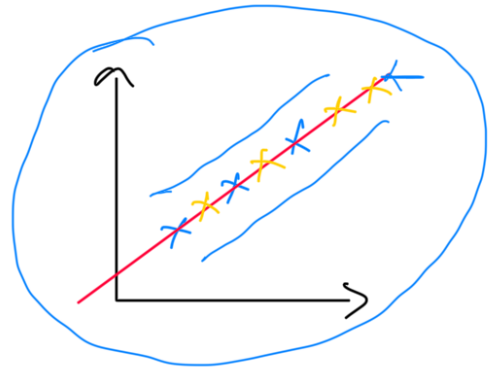
$$SSR \downarrow \rightarrow \frac{SSR}{SSM} \downarrow \rightarrow \left(1 - \frac{SSR}{SSM}\right) \uparrow \rightarrow R^2 \uparrow$$

Perfect model:

$$SSR = 0$$

$$\frac{SSR}{SSM} = 0$$

$$R^2 = 1 - 0$$



$$: R^2 = 0 \rightarrow \text{When } 1 - \frac{SSR}{SSM} = \frac{SSR}{SSM} \quad 1 - 1 = 0$$

→ our model is performing as  
same as mean line.

$$: R^2 < 0 : \underline{SSR > SSM} \rightarrow 1 - (>1) : \underline{<0} \text{ value } R^2$$

→ our model is performing worse than  
Simply predicting the mean.

$$\Rightarrow \text{Range of } R^2: (-\infty, 1]$$

⇒ definition 2: Proportion (percentage) of the total  
Variance in the target explained by  
the model.

$R^2 = 0.8$  : 80 % Variance can be explained  
by our model.